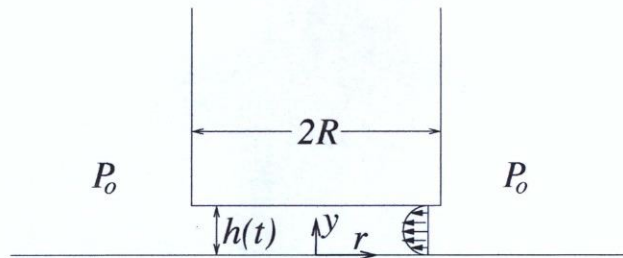


VF8

We want to lift the cylindrical body shown in the figure. A layer of oil exists between the body and the horizontal solid surface. The radial inflow caused by the lifting motion of the body is associated with the existence of underpressures in the liquid film, inducing an adhesion force to be computed here for the case of constant velocity $\dot{h} = dh/dt$.

- Show that the liquid motion in the liquid film between the body and the solid surface is dominated by viscosity for values of h much smaller than $\nu/\dot{h}$.
- Obtain the radial velocity profile v_r as a function of the radial pressure gradient $P_l = -\partial P/\partial r$.
- Using continuity, derive an equation for the pressure distribution $P(r, t)$.
- Compute the pressure distribution as well as the adhesion force acting on the body.



$$1) \quad \frac{\partial}{\partial r} (r v_r) + \frac{\partial}{\partial y} (r v_y) = 0$$

$$v_r \sim \frac{R \dot{h}}{h}$$

$$\frac{\partial (r v_y)}{\partial y} = \frac{h^2}{\nu} \frac{1}{t_0} \sim \frac{h \dot{h}}{\nu} \ll 1$$

$$\frac{\partial (r v_r)}{\partial r} = \frac{v_r h}{\nu} \frac{h}{R} \sim \frac{h \dot{h}}{\nu} \ll 1$$

$$\left. \begin{array}{l} t_0 \sim \frac{h}{\dot{h}} \\ h \ll \nu/\dot{h} \end{array} \right\} h \ll \nu/\dot{h}$$

$$2) \quad 0 = -\frac{\partial P}{\partial r} + \mu \frac{\partial^2 v_r}{\partial y^2} \Rightarrow v_r = -\frac{h^2}{2\mu} \frac{\partial P}{\partial r} \frac{y}{h} \left(1 - \frac{y}{h}\right)$$

$$3) \text{ FROM CONTINUITY } \int_0^h \left(\frac{\partial}{\partial r} (r v_r) + \frac{\partial}{\partial y} (r v_y) \right) dy = \frac{\partial}{\partial r} \int_0^h r v_r dy + r [v_y(h) - v_y(0)] = 0$$

$$4) \quad -\frac{\partial}{\partial r} \left(r \frac{h^3}{12\mu} \frac{\partial P}{\partial r} \right) + r \dot{h} = 0 \quad \left\{ \begin{array}{l} r=0: \frac{\partial P}{\partial r} = 0 \\ r=R, P=P_0 \end{array} \right. \Rightarrow \boxed{P - P_0 = \frac{3\mu}{h^3} \dot{h} (R^2 - r^2)}$$

$$\boxed{F = \int_0^R 2\pi r (P - P_0) dr \vec{e}_y = -\frac{3\pi}{2} \mu \frac{R^4}{h^3} \dot{h} \vec{e}_y}$$